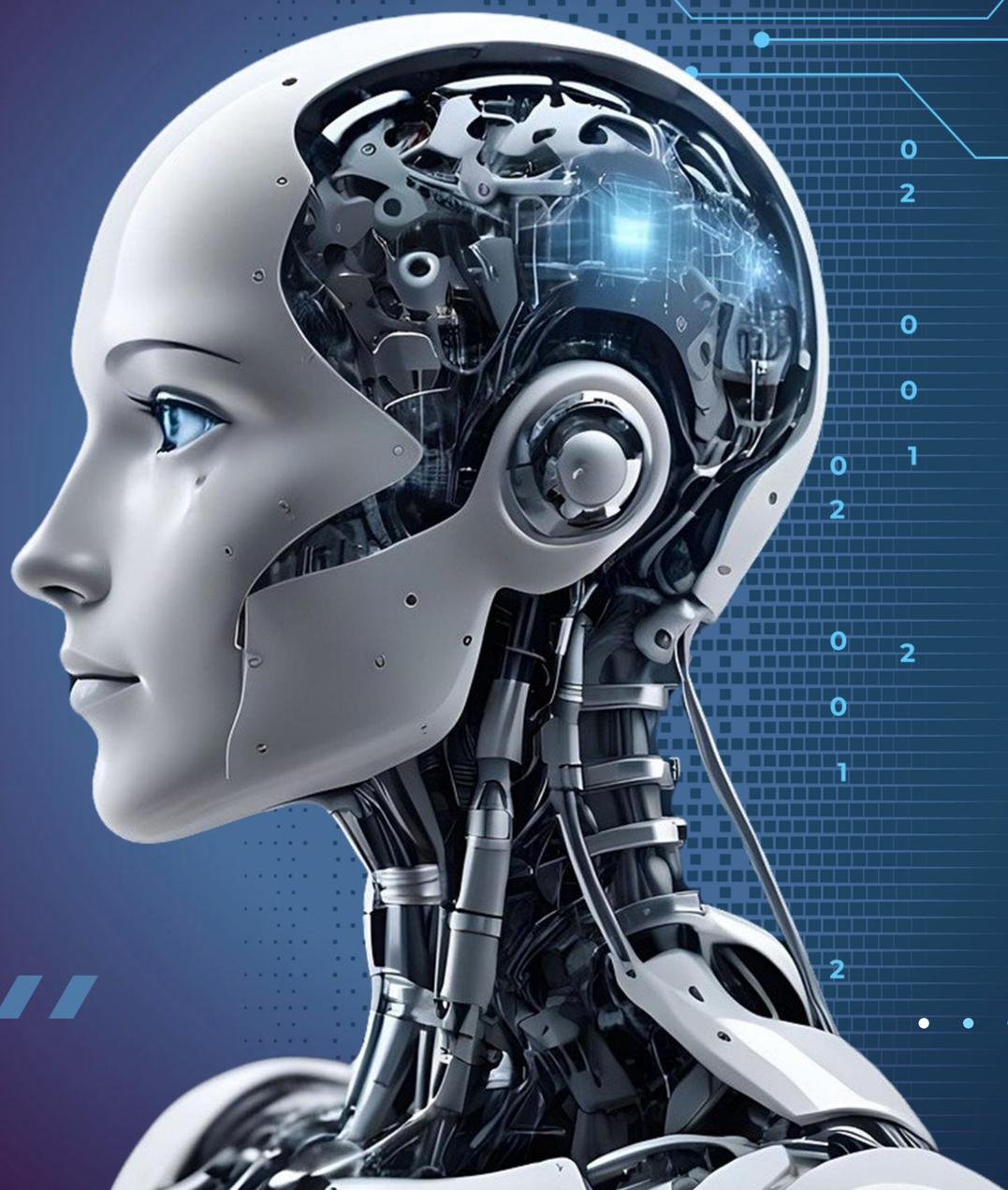
SEMINAR ON
ARTIFICIAL INTELLIGENCE

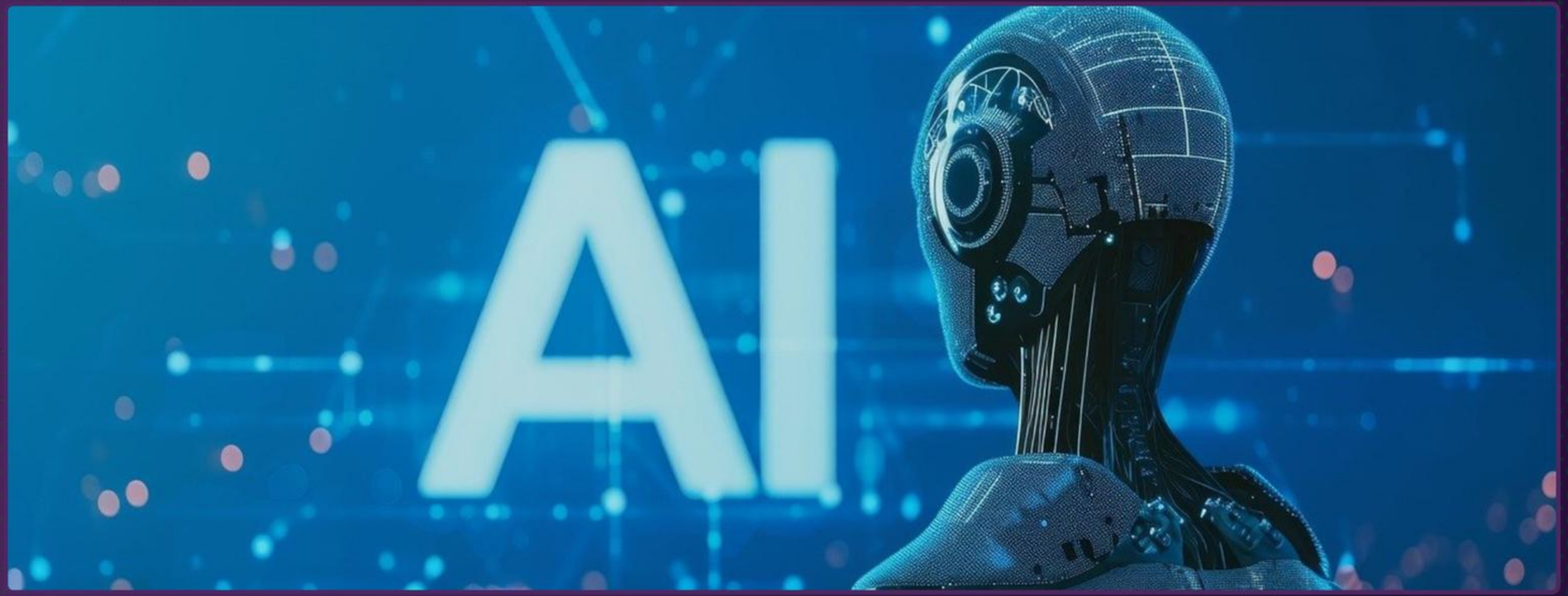
Understanding the Basics of AI and Generative AI

SYED MUHAMMAD ARSALAN SHAH BUKHARI



WELCOME TO THE SEMINAR

Syed Muhammad Arsalan Shah — AI Systems Engineer & Data Scientist | syedmuhammadarsalanshah.com



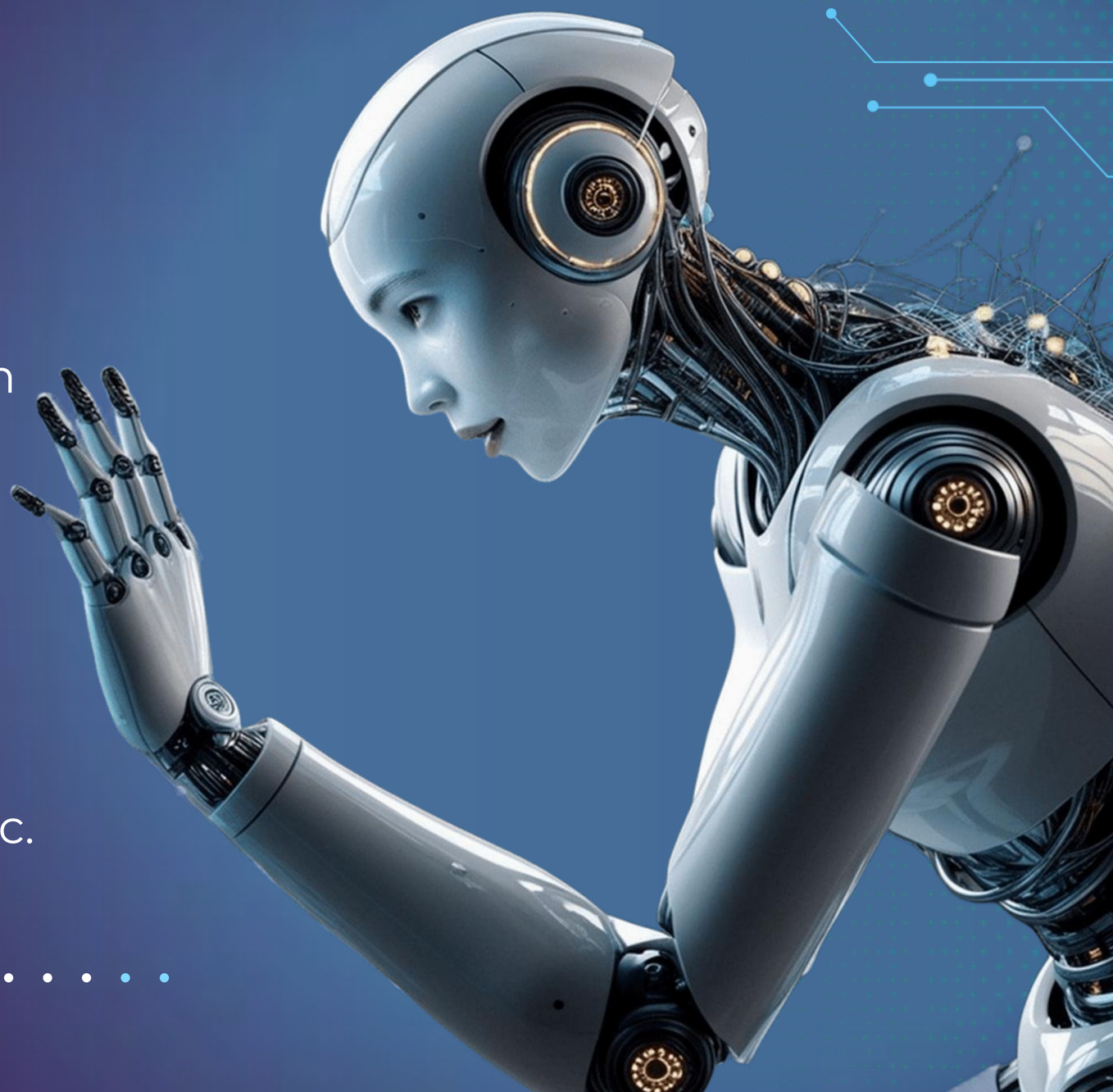
INTRODUCTION

Artificial Intelligence (AI) is the simulation of human intelligence in machines that are programmed to think, learn, reason, and make decisions.

AI systems aim to replicate human cognitive abilities

Examples of AI systems include:

- ChatGPT
- Autonomous systems by Tesla, Inc.
- AI research by OpenAI



TYPES OF ARTIFICIAL INTELLIGENCE

01.

Narrow AI (ANI)

Task-specific AI

(Current practical systems)

02.

General AI (AGI)

Human-level intelligence

(Research phase)

03.

Super AI (ASI)

Beyond human intelligence

(Theoretical concept)

USES OF AI

E-commerce

Recommendation engines

ACCOUNTABILITY AND RESPONSIBILITY

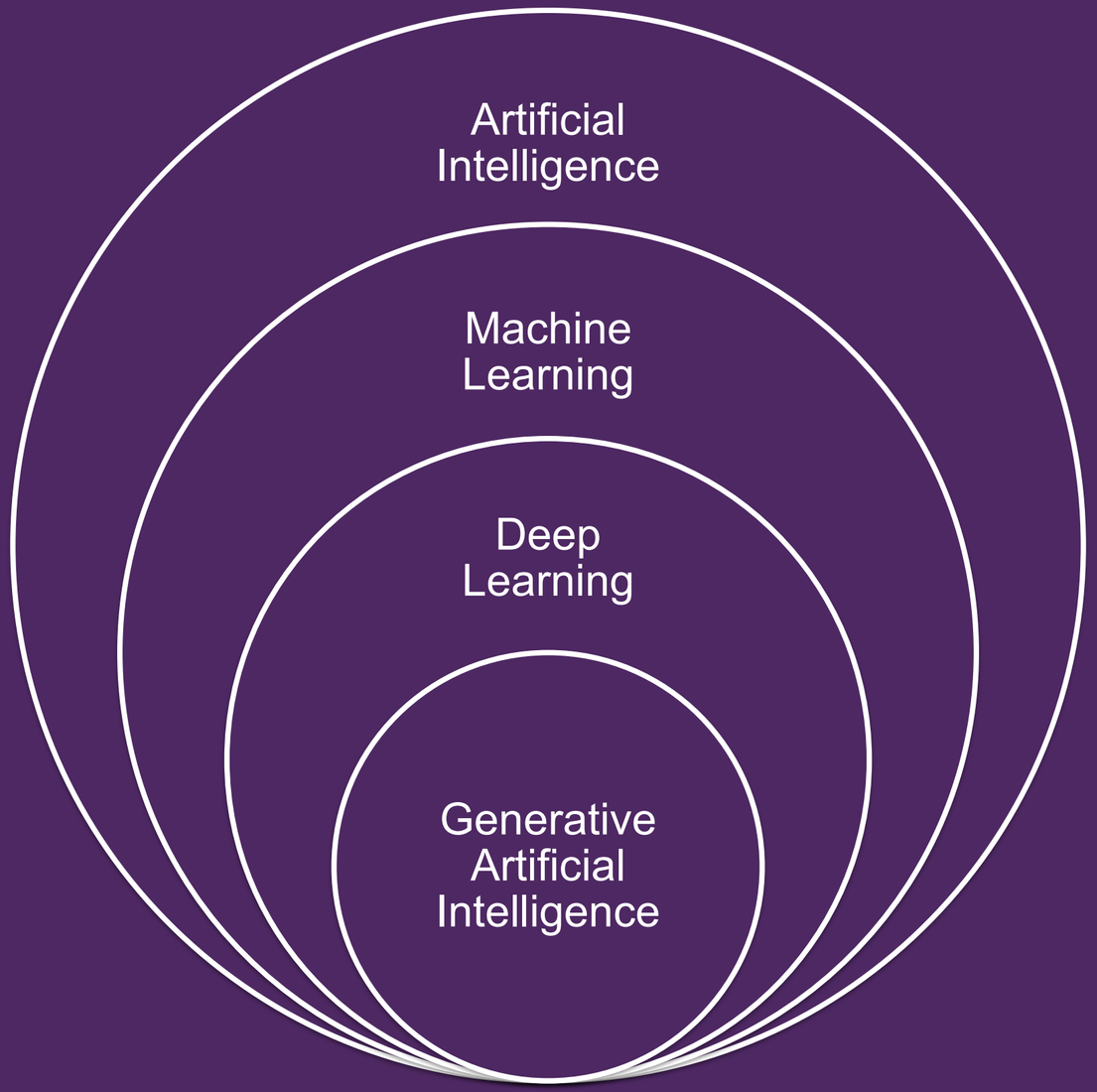
Finance → Fraud detection

Autonomous Vehicles

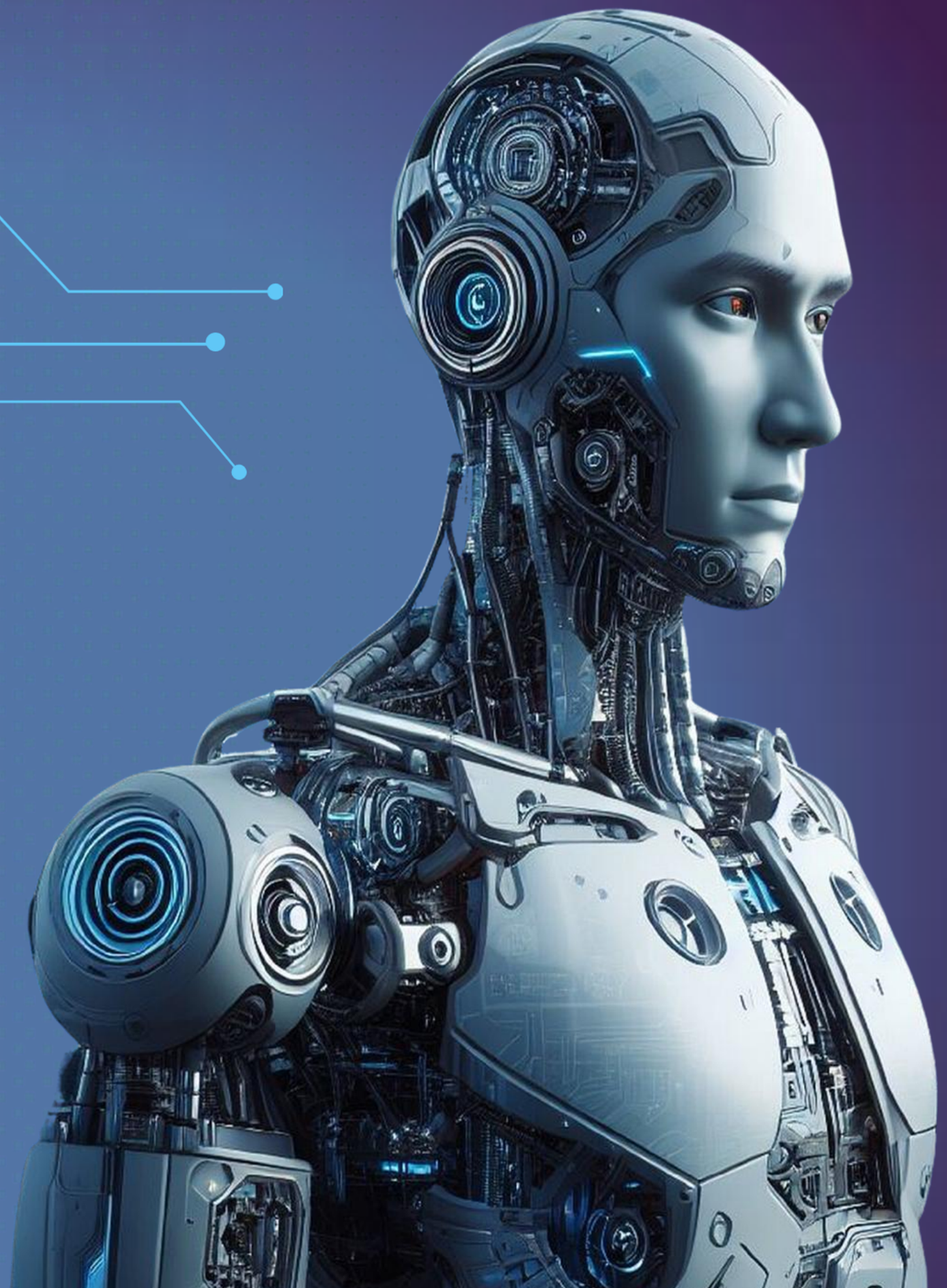
Autonomous vehicles use smart technology to drive themselves, improve safety

Healthcare

Disease prediction & diagnosis



OVERALL VENN DIAGRAM HIERARCHY



MACHINE LEARNING (ML)

Machine Learning is a subset of AI that enables systems to learn from data and improve performance automatically without being explicitly programmed.

TYPES OF MACHINE LEARNING

Supervised Learning

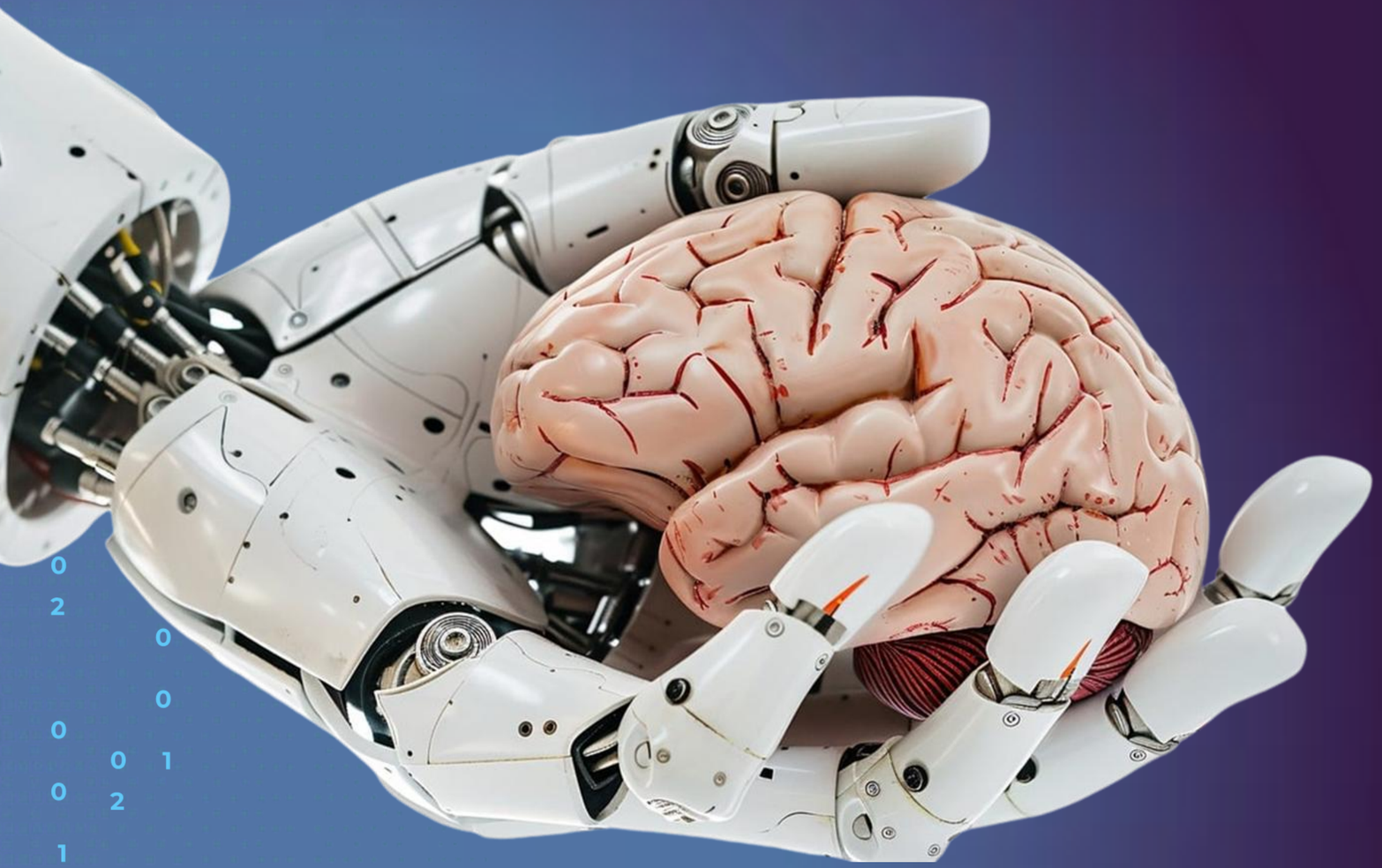
- Uses labeled data
- Regression
- Classification

Unsupervised Learning

- Uses unlabeled data
- Clustering

Reinforcement Learning

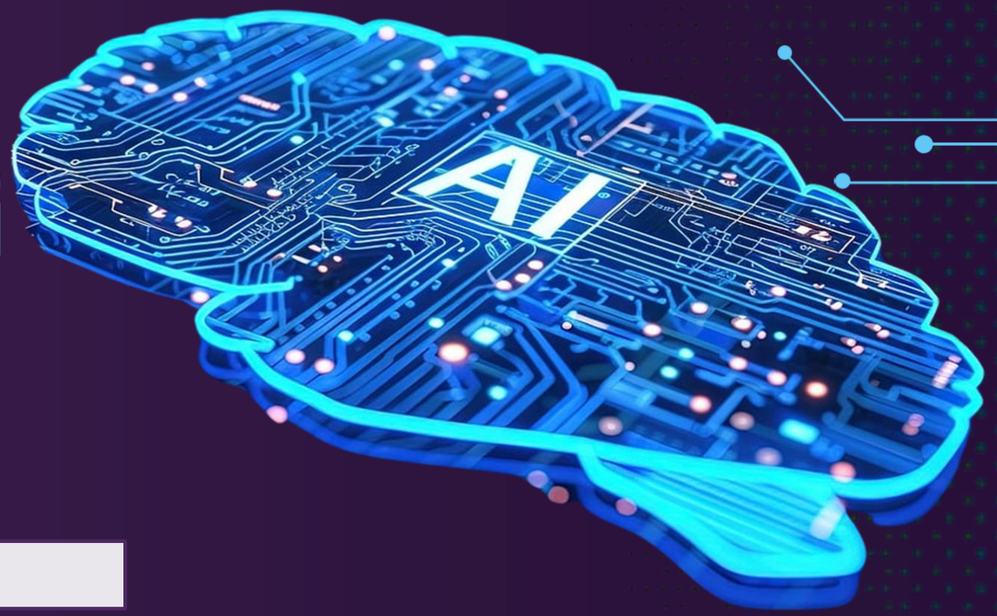
- Learning through reward and punishment
- Used in robotics & game AI



DEEP LEARNING

Deep Learning is a subset of Machine Learning based on Artificial Neural Networks with multiple hidden layers.

TYPES OF DEEP LEARNING ARCHITECTURES



- ANN (Artificial Neural Network)
- FNN (Feedforward Neural Network)
- RNN (Recurrent Neural Network)
- CNN (Convolutional Neural Network)
- Transformer Models

USES OF ML / DL

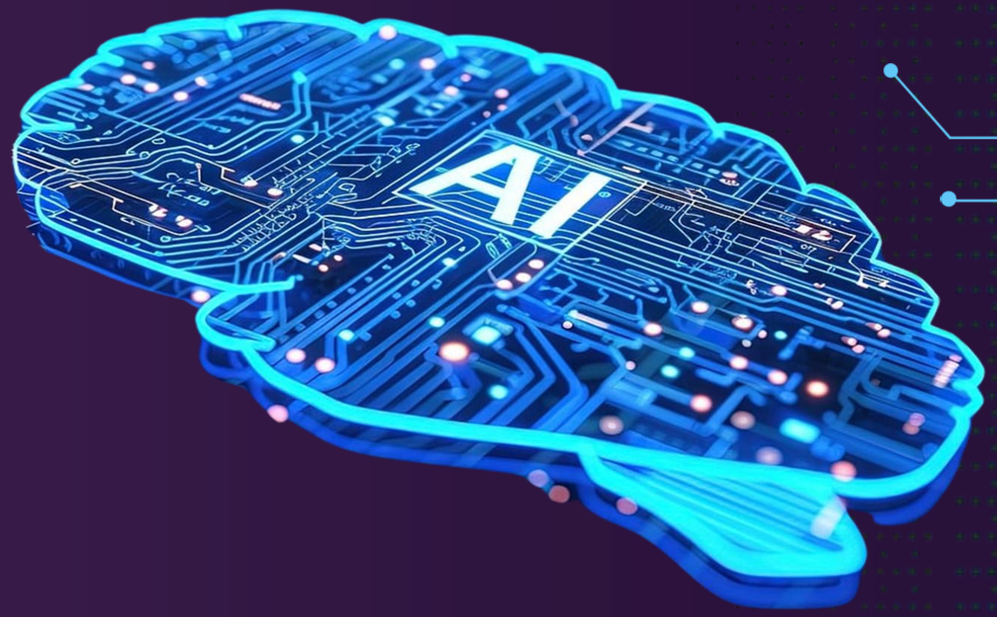
USES OF MACHINE LEARNING

- Spam detection
- Credit scoring
- Customer segmentation
- Predictive analytics
- Market forecasting

USES OF DEEP LEARNING

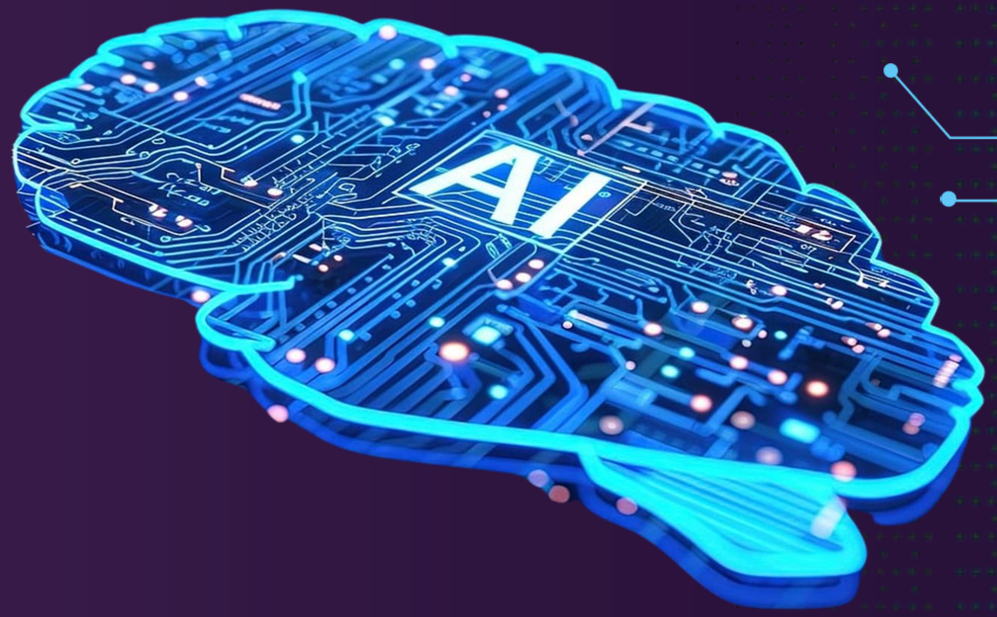
- Image recognition
- Speech recognition
- NLP systems
- Autonomous driving
- Generative AI systems.

STEPS INVOLVED IN THE AI/ML PIPELINE

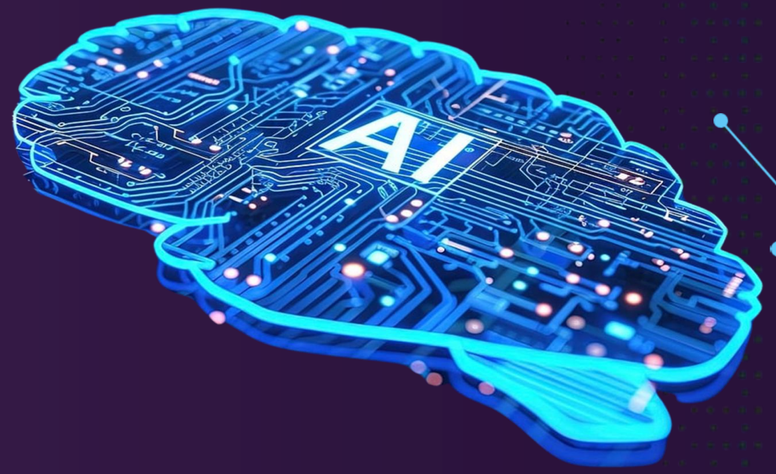


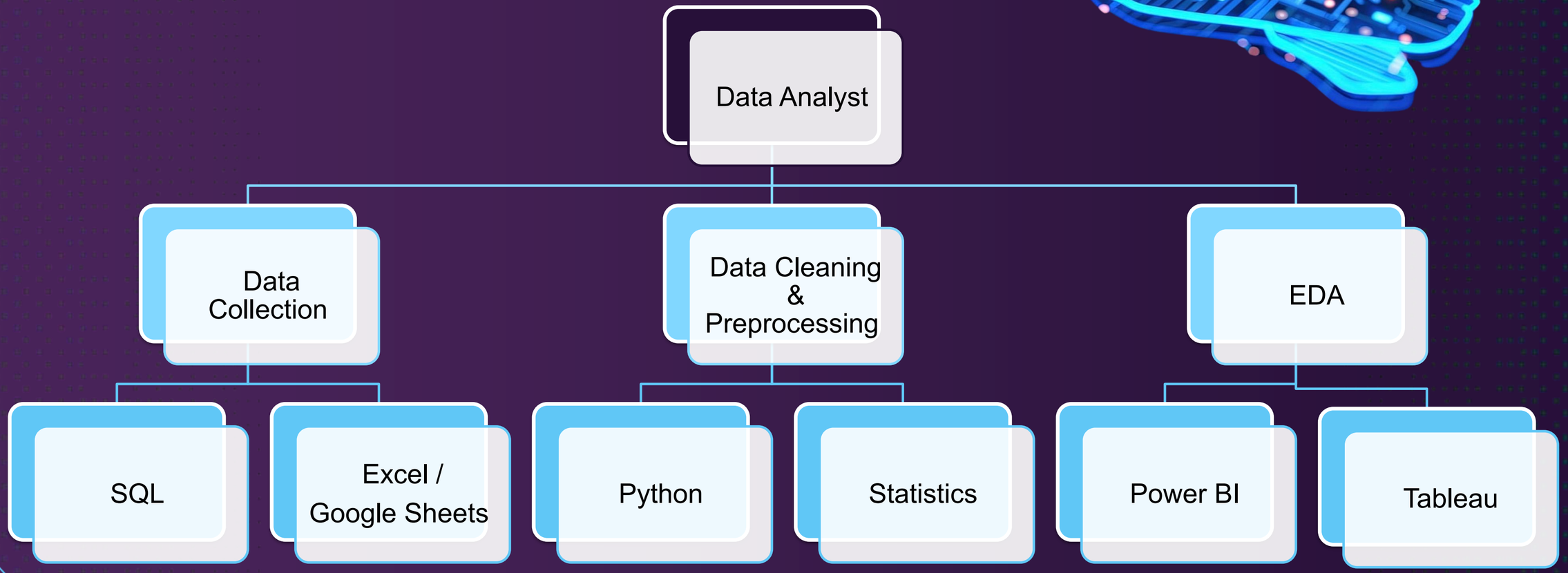
1. Data Collection
2. Data Cleaning & Preprocessing
3. Exploratory Data Analysis (EDA)
4. Feature Engineering
5. Model Selection & Training
6. Model Evaluation
7. Model Deployment

CAREER LEVEL ROADMAP SUMMARY



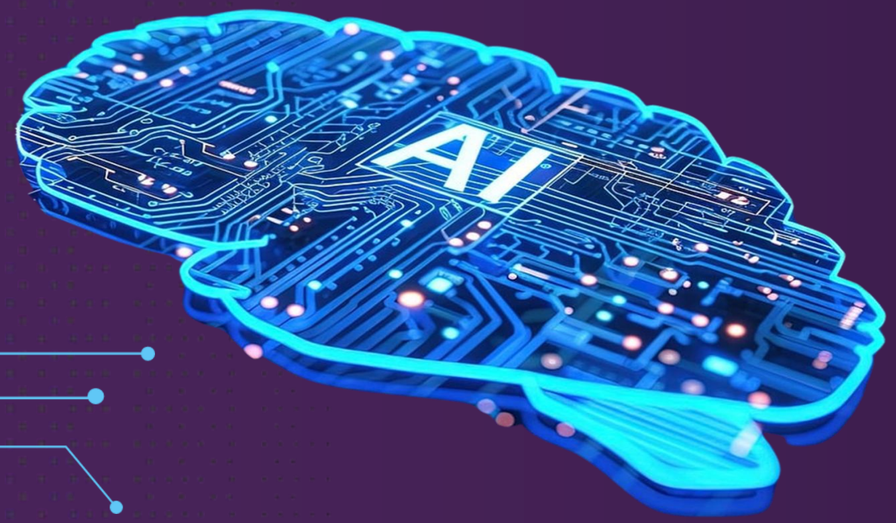
Data Analyst → Data Scientist → AI Engineer


AI


Data Analyst
Data Collection
Data Cleaning & Preprocessing
EDA
SQL
Excel / Google Sheets
Python
Statistics
Power BI
Tableau

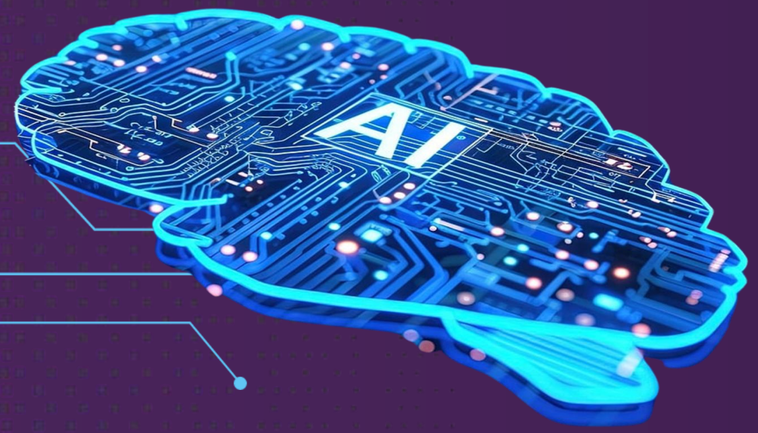
Data Scientist

- Data Collection
- SQL
- Data Cleaning & Preprocessing
  - Advanced Python
  - NumPy & Pandas
- EDA
  - Matplotlib
  - Seaborn
- Feature Engineering
- Model Selection & Training
  - Scikit-learn
    - Supervised ML
    - Unsupervised ML
    - Reinforcement Learning
  - Git & Github (additional)
  - Math
    - Statistics
    - Linear Algebra
    - Probability
    - Calculus
- Model Evaluation

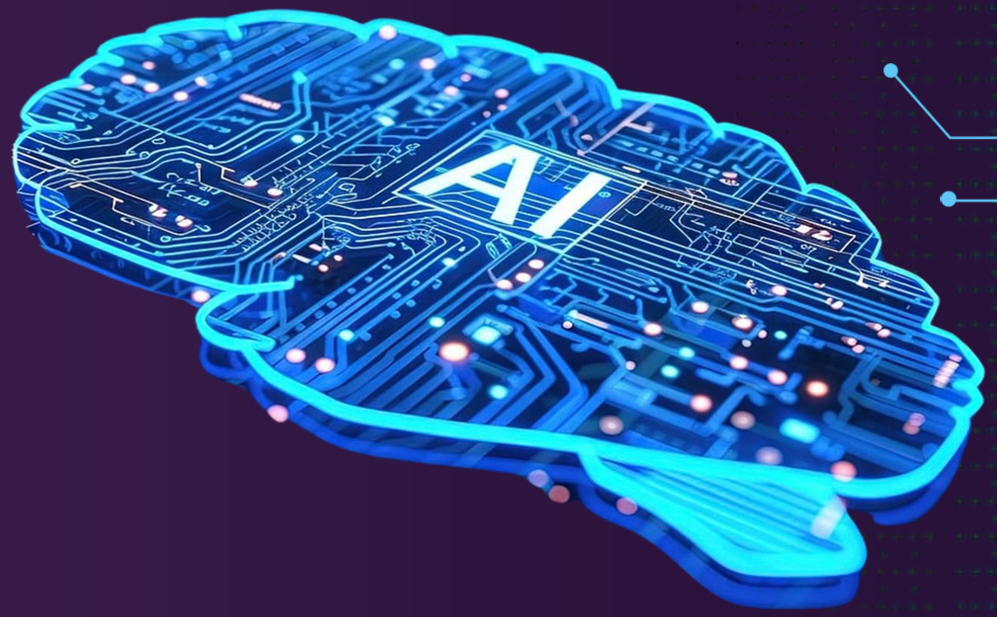


AI Engineer

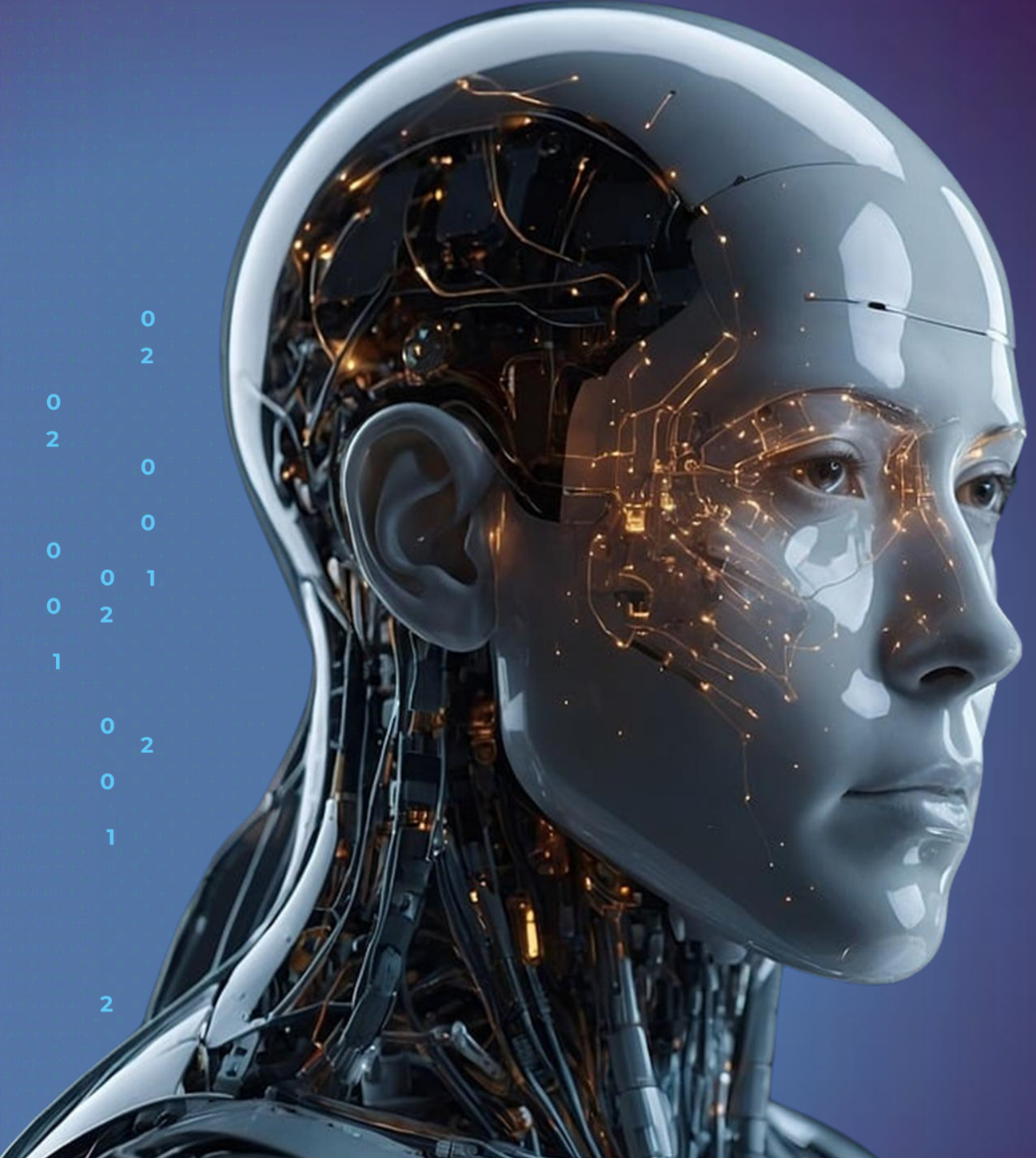
- Data Collection
  - SQL
- Data Cleaning & Preprocessing
  - Advanced Python
  - NumPy & Pandas
- EDA
  - Matplotlib
  - Seaborn
- Feature Engineering
  - Scikit-learn
    - Supervised ML
    - Unsupervised ML
    - Reinforcement Learning
  - Git & Github (additional)
- Model Selection & Training
  - Math
    - Statistics
    - Linear Algebra
    - Probability
    - Calculus
- Model Evaluation
  - Deep Learning (PyTorch or TensorFlow)
  - ANN
    - FNN
    - RNN
    - CNN
    - Transformer etc.
- Model Deployment
  - Basic Deployment
  - API Building
  - DSA (optional, asked by some companies)



CAREER PATH & SALARY OVERVIEW



Level	Focus	Outcome	Salary (Per Annum PKR Approx.)
Data Analyst	Insights	Business Intelligence	Rs 1.2M – Rs 1.8M
Data Scientist	Prediction	Smart Decision Systems	Rs 1.5M – Rs 3.3M
AI Engineer	Intelligent Systems	Automation & AI Products	Rs 2M – Rs 4M



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جزاک اللہ خیرا

افراد کے ہاتھوں میں ہے اقوام کی تقدیر
ہر فرد ہے ملّت کے مقدّر کا ستارا